

Real Analysis I

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Lecture 31

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0.1 Properties of Integral

Example. 1. When $g(x) = x$, the R-S integral is called [\[\[Riemann Integral \]\]](#) .

2. Let

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

be the *Dirichet's function*.

Claim. f not Riemann integrable on any $[a, b], a < b$.

Proof. Let P be any partition of $[a, b]$. By density of rationals (and irrationals),

$$m_i = \inf_{[x_{i-1}, x_i]} f = 0 \quad \text{for any } i = 1, \dots, N.$$

$$M_i = \sup_{[x_{i-1}, x_i]} f = 1 \quad \text{for any } i = 1, \dots, N.$$

Hence $L(P, f) = 0$, $U(P, f) = \sum_{i=1}^N 1\Delta x_i = b - a$. Hence the upper and lower sums do not match. Hence not integrable in Darboux sense. $\square$

3. Let

$$g(x) = \begin{cases} 0 & a \leq x \leq c \\ 1 & c < x \leq b \end{cases}, c \in (a, b).$$

Let P be a partition containing c . Let i be such that $c = x_{i-1}$. Then note: $\Delta g_k = 0$ unless $k = i$. Moreover, $\Delta g_i = 1$. Therefore,

$$S(P, f, g) = \sum_{k=1}^N f(\xi_k) \Delta g_k \quad \xi_k \in [x_{k-1}, x_k].$$

$$S(P, f, g) = f(\xi_i).$$

$\int_a^b f dg$ exists iff $\lim_{x \rightarrow c+} f(x) = f(c)$ (continuity from the right). We have $\int_a^b f dg = f(c)$.

Remark. This allows us to treat series as integrals.

◇

Theorem 1 (Change of Variable). Suppose $g(x)$ monotone increasing and **continuously differentiable**. Let $f(x)$ be a bounded function. Then $\int_a^b f dg$ exists iff $\int_a^b f g' dx$ exists, and

$$\int_a^b f dg = \int_a^b f g' dx.$$

Proof. Since g' continuous on $[a, b]$, g' uniformly continuous on $[a, b]$. Hence for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in [a, b]$,

$$|x - y| < \delta \Rightarrow |g'(x) - g'(y)| < \varepsilon.$$

Now pick P_ε be a partition such that $\Delta x_i < \delta$. Then for any P finer than P_ε we have $\Delta x_i < \delta$ for P as well. Now

$$S(P, f, g) = \sum_{i=1}^N f(\xi_i) (g(x_i) - g(x_{i-1})).$$

$$S(P, f g') = \sum_{i=1}^N f(\xi_i) g'(\xi_i) (x_i - x_{i-1}).$$

Apply mean value theorem, there exists $\zeta \in [x_{i-1}, x_i]$ such that

$$f(\xi_i) (g(x_i) - g(x_{i-1})) = f(\xi_i) g'(\zeta_i) (x_i - x_{i-1}).$$

Now

$$|S(P, f, g) - S(P, f g')| \leq \sum_{i=1}^N |f(\xi_i)| |g'(\zeta_i) - g'(\xi_i)| \Delta x_i$$

$$\leq \|f\| \varepsilon (b - a).$$

Now if one of the integrals exists, then this statement implies that the other integral must exist. Hence the theorem holds. $\square$

Proposition 1. (Properties of [R-S integral](#)) Suppose $f_1, f_2 : [a, b] \rightarrow \mathbb{R}$ bounded, and g monotone increase. Then if $\int_a^b f_1 dg$ and $\int_a^b f_2 dg$ exist, then

- (Linearity) For $\alpha_1, \alpha_2 \in \mathbb{R}$,

$$\int_a^b (\alpha_1 f_1 + \alpha_2 f_2) dg = \alpha_1 \int_a^b f_1 dg + \alpha_2 \int_a^b f_2 dg.$$

- (Linearity in integrator) If $\beta_1, \beta_2 \geq 0$, then

$$\int_a^b f d(\beta_1 g_1 + \beta_2 g_2) = \beta_1 \int_a^b f dg_1 + \beta_2 \int_a^b f dg_2.$$

Proof. • (Linearity)

- (Linearity in integrator) Riemann Sum is linear.

□

Proposition 2. (Additivity of [R-S integral](#)) Let $f : [a, b] \rightarrow \mathbb{R}$ bounded and g monotone increase, $c \in [a, b]$. Then $\int_a^b f dg$ exists iff $\int_a^c f dg$ and $\int_c^b f dg$ exist, and then

$$\int_a^b f dg = \int_a^c f dg + \int_c^b f dg.$$

Proof. Let $I = \int_a^b f dg$. Without loss of generality we consider only partitions that contain c . For any $\varepsilon > 0$, there exists P_ε such that

$$|S(P, f, g) - I| < \varepsilon \quad \text{if } P > P_\varepsilon.$$

Assume existence of integral, then for $P > P_\varepsilon$,

$$|U(P, f, g) - L(P, f, g)| \leq 2\varepsilon.$$

For $P > P_\varepsilon$, P contains $c = x_M$. Then $P' : a = x_0 < x_1 < \dots < x_M = c$ is a partition of $[a, c]$, and $P'' : c = x_M < x_{M+1} < \dots < x_N = b$ is a partition of $[c, b]$. Now

$$\begin{aligned} |U(P, f, g) - L(P, f, g)| &= \sum_{i=1}^N (M_i - m_i) \Delta g_i \\ &\geq \sum_{i=1}^M (M_i - m_i) \Delta g_i \\ &= U(P', f, g) - L(P', f, g). \end{aligned}$$

Hence for $P > P_\varepsilon$,

$$|U(P', f, g) - L(P', f, g)| \leq 2\varepsilon.$$

Hence

$$\left| \overline{\int_a^c f dg} - \underline{\int_a^c f dg} \right| \leq 2\varepsilon.$$

Hence the integral exists. Similarly $\int_c^b f dg$ exists.

On the other direction, suppose both $\int_a^c f dg$ and $\int_c^b f dg$ exists. Then we can put two partitions together in a similar way, and conclude that for fine P ,

$$\left| S(P, f, g) - \left(\int_a^c f dg + \int_c^b f dg \right) \right| \leq 2\varepsilon.$$

Now to verify additive identity,

$$S(P, f, g) = S(P', f, g) + S(P'', f, g).$$

Hence by taking fine partitions, each is ε -close to the corresponding integral. So

$$\left| \int_a^b f dg - \left(\int_a^c f dg + \int_c^b f dg \right) \right| \leq 3\varepsilon \rightarrow 0.$$

□

Lecture 32

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Theorem 2 (Mean Value Theorem). If f is continuous on $[a, b]$, g monotone increase on $[a, b]$, then there is $c \in [a, b]$ such that

$$\int_a^b f dg = f(c) [g(b) - g(a)].$$

Remark. In this theorem c is in the closed interval, and we cannot restrict it to the open interval. For example, let g be constantly zero on $[a, b]$ and 1 on $[b, \infty)$. Then $\int_a^b f dg = f(b)(g(b) - g(a))$.

Theorem 3 (Differentiation of Integral). Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$ and continuous at $c \in [a, b]$. Let

$$F(x) = \int_a^x f(t) dt.$$

Then $F'(c) = f(c)$.

Definition 1 (Antiderivative). We say that G is an *antiderivative* of f on $[a, b]$ if

1. G continuous on $[a, b]$
2. $G'(x) = f(x)$ on (a, b)

Corollary 1 (Fundamental Theorem of Calculus). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Let G be an antiderivative of f on $[a, b]$. Then

$$\int_a^b f(x)dx = G(b) - G(a).$$

Also called [\[\[Newton-Leibniz Formula \]\]](#).

Proof. Let $F(x) = \int_a^x f(t)dt$, $x \in [a, b]$. We claim that F is antiderivative of f . (2) follows from differentiation theorem. (1) follows from the fact that as

$$|F(y) - F(x)| = \left| \int_x^y f(t)dt \right| \leq M|x - y|.$$

hence F Lipschitz continuous, hence continuous on $[a, b]$.

Then look at

$$H(x) = G(x) - F(x).$$

We have

1. H continuous on $[a, b]$
2. $H'(x) = 0$ for every $x \in (a, b)$

Hence H is constant on $[a, b]$. Hence $G(x) = F(x) + C$ for all $x \in [a, b]$, for some $C \in \mathbb{R}$. Now

$$G(b) - G(a) = (F(b) + C) - (F(a) + C) = F(b) - F(a) = F(b) = \int_a^b f(t)dt.$$

□

Theorem 4 (Integration By Parts). Suppose f, g continuously differentiable on $[a, b]$. Then

$$\int_a^b fg'dx = fg|_a^b - \int_a^b gf'dx.$$

Proof. From product rule, $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$. We can integrate this since $(fg)'$ is continuous. Now by Newton-Leibniz,

$$\int_a^b (fg)'dx = f(b)g(b) - f(a)g(a).$$

On the other hand

$$\int_a^b (fg)'dx = \int_a^b fg'dx + \int_a^b gf'dx.$$

□

Theorem 5. If f bounded on $[a, b]$, g monotone increase and continuously differentiable on $[a, b]$, then

$$\int_a^b f dg \text{ exists} \Leftrightarrow \int_a^b f g' dx \text{ exists.}$$

and the two integrals are equal.

Theorem 6. Suppose f, g monotone increase on $[a, b]$, then

$$\int_a^b f dg \text{ exists} \Leftrightarrow \int_a^b g df \text{ exists.}$$

and then

$$\int_a^b f dg = fg|_a^b - \int_a^b g df.$$

Proof. By Riemann Criterion for integrability, $\int_a^b f dg$ exists iff for any $\varepsilon > 0$ there exists partition P_ε such that

$$\sum_{i=1}^N (M_i(f) - m_i(f)) \Delta g_i < \varepsilon \quad .$$

where

$$m_i(f) = \inf_{[x_{i-1}, x_i]} f = f(x_{i-1}).$$

$$M_i(f) = \sup_{[x_{i-1}, x_i]} f = f(x_i).$$

Hence

$$\sum_{i=1}^N (M_i(f) - m_i(f)) \Delta g_i = \sum_{i=1}^N \Delta f_i \Delta g_i = \sum_{i=1}^N (M_i(g) - m_i(g)) \Delta f_i.$$

Hence the existence criterion for the two integrals are equivalent.

Let P be any partition of $[a, b]$ such that

$$U(P, f, g) = \sum_{i=1}^N f(x_i)[g(x_i) - g(x_{i-1})].$$

$$L(P, g, f) = \sum_{i=1}^N g(x_{i-1})[f(x_i) - f(x_{i-1})].$$

Now

$$\begin{aligned} U(P, f, g) + L(P, g, f) &= \sum_{i=1}^N [f(x_i)g(x_i) - f(x_{i-1})g(x_{i-1})] \\ &= f(b)g(b) - f(a)g(a). \end{aligned}$$

Hence

$$\int_a^b f dg + \int_a^b g df = f(b)g(b) - f(a)g(a).$$

□

Theorem 7 (Change of Variables). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and $\varphi : [\alpha, \beta] \rightarrow [a, b]$ be continuously differentiable and such that

$$\varphi(\alpha) = a, \varphi(\beta) = b.$$

Then

$$\int_a^b f(x)dx = \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt.$$

Proof. Let

$$F(x) = \int_a^x f(\xi)d\xi.$$

and

$$\Phi(t) = \int_\alpha^t f(\varphi(\tau))\varphi'(\tau)d\tau.$$

We look at $F(\varphi(\tau)), \tau \in [\alpha, \beta]$

1. $F(\varphi(\tau))$ continuous on $[\alpha, \beta]$
2. $(F(\varphi(\tau)))' = F'(\varphi(\tau))\varphi'(\tau),$

so $F(\varphi(\tau))$ is an antiderivative of $f(\varphi(t))\varphi'(t)$ on $[\alpha, \beta]$.

By the Newton-Leibniz Formula,

$$\int_\alpha^\beta f(\varphi(t))\varphi'(t)dt = F(\varphi(\beta)) - F(\varphi(\alpha)) = F(b) - F(a) = \int_a^b f(x)dx.$$

□

Theorem 8. Suppose $f : [a, b] \rightarrow \mathbb{R}$ bounded and $\varphi : [\alpha, \beta] \rightarrow [a, b]$ monotone increase, and φ is a bijection. Then

$$\int_a^b f(x)dx \text{ exists} \Leftrightarrow \int_\alpha^\beta f \circ \varphi d\varphi \text{ exists.}$$

and the integrals are equal.

Proof. Let P be a partition of $[\alpha, \beta]$. Denote $P : \alpha = t_0 < t_1 < \dots < t_N = \beta$. Let $\varphi(P) : a = x_0 < x_1 = \varphi(t_1) < x_2 = \varphi(t_2) < \dots < x_N = \beta$.

Now

$$\begin{aligned} U(P, f, \varphi) &= U(\varphi(P), f, x) \\ L(P, f, \varphi) &= L(\varphi(P), f, x). \end{aligned}$$

This finishes the proof.

□

0.2 Convergence and Integral

In general, if $f_n : [a, b] \rightarrow \mathbb{R}$ are R-S integrable w.r.t g monotone increase on $[a, b]$.

$f_n \rightarrow f$ pointwise, does not imply f R-S integrable, and even if the limit exists, the integrals are not necessarily equal.

Example. Let f be the Dirichlet function

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}.$$

We enumerate the rational numbers $\mathbb{Q} = \{r_n : n \in \mathbb{N}\}$. Let

$$f_n(x) = \begin{cases} 1 & x = r_1, r_2, \dots, r_n \\ 0 & \text{otherwise} \end{cases}.$$

f_n has finitely many discontinuities, so f_n is R-S integrable. Also, $f_n \rightarrow f$ pointwise. But f not R-S integrable. $\diamond$

Example. Let f_n be piecewise linear such that

$$f_n(0) = 0, f_n\left(\frac{1}{n}\right) = n, f_n\left(\frac{2}{n}\right) = 0.$$

Then $f_n \rightarrow 0$ pointwise, but $\int_0^1 f_n dx = 1$. $\diamond$

Theorem 9 (Bounded Convergence Theorem). Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are uniformly bounded on $[a, b]$, i.e. there exists $B \geq 0$ such that

$$|f_n(x)| \leq B \text{ for all } x \in [a, b], n \in \mathbb{N}.$$

If f_n are R-S integrable wrt g and $f_n(x) \rightarrow f(x)$ pointwise, then f is also R-S integrable and

$$\int_a^b f_n dg \rightarrow \int_a^b f dg.$$

This theorem is easy to prove in the Lebesgue integration theory, but very difficult to prove in RS integration theory. Hence we omit the proof.

Theorem 10. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be R-S integrable wrt g , g monotone increase on $[a, b]$, and $f_n \rightarrow f$ uniformly on $[a, b]$. Then f is R-S integrable wrt g on $[a, b]$, and $\int_a^b f_n dg \rightarrow \int_a^b f dg$.

Proof. Suppose f is integrable, then

$$\left| \int_a^b f_n dg - \int_a^b f dg \right| \leq \int_a^b |f_n - f| dg \leq \varepsilon \int_a^b dg = \varepsilon(g(b) - g(a)).$$

Now we verify integrability of f . $\square$

For any ε , we have for sufficiently fine partition that

$$|U(P, f_n, g) - U(P, f, g)| \leq \varepsilon (g(b) - g(a)).$$

Hence for $n \geq K_\varepsilon$,

$$\left| \int_a^b f_n dg - \int_a^b f dg \right| \leq \varepsilon.$$

Symmetrically we have the result for lower integral. But since f_n integrable, we have

$$\left| \int_a^b f dg - \int_a^b f dg \right| \leq 2\varepsilon.$$

This proves f is RS integrable.

Lecture 33: Series

Wed 30 Nov 2022 11:30

1 Series

1.1 Definitions

Definition 2. (Series of Numbers) Define the *partial sum*

$$S_n = \sum_{k=1}^n x_k.$$

We say

$$S = \sum_{k=1}^{\infty} x_k : \Leftrightarrow S_n \rightarrow S \text{ as } n \rightarrow \infty.$$

Definition 3 (Absolutely Convergent). We say a series $\sum_{k=1}^{\infty} x_k$ is *absolutely convergent* if $\sum_{k=1}^{\infty} |x_k|$.

Lemma 1. Let $\sum_{k=1}^{\infty} x_n$ be a series with $x_n \geq 0$. Then it is convergent iff there exists M such that $S_n \leq M$ for all n . Then $\sum_{k=1}^{\infty} x_n \leq M$.

Proof. By Monotone Convergence Theorem applied to S_n . □

Lemma 2 (Cauchy Criterion). $\sum x_n$ converges iff for any $\varepsilon > 0$ there is M_ε such that $\sum_{k=n}^m x_k$ if $n \geq m \geq M_\varepsilon$.

Proof. S_n converges iff S_n is cauchy sequence. □

Proposition 3. Absolute convergence implies convergence.

Proof. Use triangle inequality, prove that $\sum x_n$ is Cauchy. $\square$

Definition 4 (Rearrangement of Series). Take $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. The series

$$\sum_{n=1}^{\infty} x_{\varphi(n)}$$

is called a *rearrangement* of $\sum_{n=1}^{\infty} x_n$.

Theorem 11. If $x_n \geq 0$, $\sum_{n=1}^{\infty} x_n < \infty$, then $\sum_{n=1}^{\infty} x_{\varphi(n)} < \infty$ and $\sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} x_{\varphi(n)}$

Proof. Let $S'_n = X_{\varphi(1)} + \dots + X_{\varphi(n)}$. Let $N = \max\{\varphi(1), \dots, \varphi(n)\}$. Hence for any n ,

$$S'_n \leq S_N \leq M.$$

Thus $\sum_{n=1}^{\infty} x_{\varphi(n)} = \lim_{n \rightarrow \infty} S'_n \leq M$. $\square$

Theorem 12. If $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is bijection, then $\sum x_n$ absolutely convergent iff $\sum x_{\varphi(n)}$ absolutely convergent, and then

$$\sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} x_{\varphi(n)}.$$

Proof. Convergence follows from the proposition above. To prove the sums are equal, split into positive and negative parts. $\square$

Definition 5 (Conditionally Convergent). $\sum x_n$ is convergent, but not absolutely.

Equivalently, both $\sum_{n=1}^{\infty} x_n^+$ and $\sum_{n=1}^{\infty} x_n^-$ are infinite.

Theorem 13. If $\sum_{n=1}^{\infty} x_n$ converges conditionally, then for each real number x there exists $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\sum_{n=1}^{\infty} x_{\varphi(n)} = x.$$

Proposition 4. $\sum a_n$ convergent implies $a_n \rightarrow 0$.

Proof. $a_n = S_n - S_{n-1} \rightarrow S - S = 0$ □

Example. 1. (Geometric series)

2. (p -series)

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

If $p \leq 0$, then $\frac{1}{n^p} \geq 1 \not\rightarrow 0$ divergent. If $p > 0$, $\frac{1}{n^p} \rightarrow 0$. But is the series it convergent?

Apply *Cauchy's condensation method*:

$$S_{2^k} \geq 1 + \frac{1}{2} \left(\frac{1}{2^{p-1} + \frac{1}{(2^{p-1})^2}} + \dots + \frac{1}{(2^{p-1})^k} \right).$$

$$S_{2^k} \geq 1 + \frac{1}{2} (a + a^2 + \dots + a^k) \quad a = \frac{1}{2^{p-1}}.$$

So the series diverges if $p \leq 1$.

◇

Theorem 14 (Cauchy's Condensation Test). Let $a_n \geq 0$, a_n strictly decreasing. Then $\sum a_n < \infty$ iff $\sum 2^k a_{2^k} < \infty$.

Theorem 15 (Comparison Theorem). Suppose $\sum x_n$, $\sum y_n$, $x_n, y_n \geq 0$ are such that $x_n \leq y_n$ for $n \geq N$. Then

$$\sum y_n < \infty \Rightarrow \sum x_n < \infty.$$

Also

$$\sum x_n = \infty \Rightarrow \sum y_n = \infty.$$

Proof. work with partial sums. □

Theorem 16 (Limit Comparison Theorem). Suppose $\limsup \frac{x_n}{y_n} < \infty$, then

$$\sum y_n < \infty \Rightarrow \sum x_n < \infty.$$

Suppose $\liminf \frac{x_n}{y_n} > 0$ then

$$\sum y_n = \infty \Rightarrow \sum x_n = \infty.$$

Theorem 17 (Root Test). Let $x_n \geq 0$ and

$$\rho = \limsup_{n \rightarrow \infty} |x_n|^{\frac{1}{n}}.$$

Then

1. If $\rho < 1$ then $\sum x_n$ absolutely convergent
2. If $\rho > 1$ then $\sum x_n$ divergent.
3. Otherwise the test is conclusive.

Proof. 1. Choose $\rho < \tilde{\rho} < 1$. Then there exists N such that for $n \geq N$,

$$|x_n|^{\frac{1}{n}} \leq \tilde{\rho}.$$

Hence $|x_n| \leq \tilde{\rho}^n$. Now we can apply comparison test.

2. Pick subsequence such that $|x_{n_k}|^{\frac{1}{n_k}} \geq 1$, so $x_n \not\rightarrow 0$.

□

Lecture 34: Rearrangement of Series

Fri 02 Dec 2022 11:32

- Convergence and sum is preserved under rearrangement when the convergence is absolute.
- If a sum converges but not absolutely, then both positive and negative parts must diverge.

Lecture 35

Mon 05 Dec 2022 11:29

Theorem 18 (Ratio Test). Suppose $x_n \neq 0$, and for $n \geq N$,

1. If $\limsup \frac{|x_{n+1}|}{|x_n|} < 1$, then $\sum |x_n| < \infty$
2. If $\liminf \frac{|x_{n+1}|}{|x_n|} > 1$, then $\sum |x_n|$ diverges
3. Otherwise the test is inconclusive.

Proof. Recall, from homework,

- $\limsup |x_n|^{\frac{1}{n}} \leq \limsup \frac{|x_{n+1}|}{|x_n|}.$
- $\liminf |x_n|^{\frac{1}{n}} \geq \liminf \frac{|x_{n+1}|}{|x_n|}.$

□

Theorem 19 (Dirichlet Test for conditional convergence). Let $\sum_{n=1}^{\infty} x_n y_n$ be such that

1. x_n monotonely decrease to zero
2. Partial sums $s_n = \sum_{k=1}^n y_k$ are bounded.

Then $\sum_{n=1}^{\infty} x_n y_n$ is convergent.

Proof. By the Cauchy Criterion, we consider

$$\begin{aligned} & x_{m+1}y_{m+1} + \dots + x_n y_n \\ &= x_{m+1}(s_{m+1} - s_m) + x_{m+2}(s_{m+2} - s_{m+1}) + \dots + x_n(s_n - s_{n-1}) \\ &= x_n s_n - x_{m+1} s_m + (x_{m+1} - x_{m+2})s_{m+1} + \dots + (x_{n-1} - x_n)s_{n-1} \\ &= x_n s_n - x_{m+1} s_m + \sum_{k=m+1}^n (x_{k-1} - x_k) s_{k-1}. \end{aligned}$$

Using this, we can bound the sum by

$$\left| \sum_{k=m+1}^n x_k y_k \right| \leq B|x_n| + |x_{m+1}| + B|x_m - x_n| \leq 4\varepsilon B.$$

This part can be generalized since the replacement of S_n by B is uniform. $\square$

As a particular case, we have convergence condition for alternating series.

Theorem 20 (Alternating Series Test). If x_n monotone decrease to zero, then $\sum (-1)^n x_n$ converges.

Example. (p310 on textbook) Consider $\sum_{n=1}^{\infty} \frac{\cos nx}{n}$. Take $x_n = \frac{1}{n}$ monotone increase to 0, then

$$\begin{aligned} s_n &= \cos x + \cos 2x + \dots + \cos nx \\ &= \frac{1}{\sin \frac{x}{2}} \left(\sin \frac{x}{2} \cos x + \sin \frac{x}{2} \cos 2x + \dots + \sin \frac{x}{2} \cos nx \right) \\ &= \frac{1}{\sin \frac{x}{2}} \left(\left(\sin \frac{3}{2}x - \sin \frac{x}{2} \right) + \left(\sin \frac{5}{2}x - \sin \frac{3}{2}x \right) + \dots + \left(\sin(n + \frac{1}{2})x - \sin(n - \frac{1}{2})x \right) \right) \\ &= \frac{\sin(n + \frac{1}{2})x - \sin \frac{x}{2}}{2 \sin \frac{x}{2}}. \end{aligned}$$

Hence $|s_n| \leq \frac{1}{|\sin \frac{x}{2}|}$. So, if $\sin \frac{x}{2} \neq 0$, then s_n are bounded, so the sum converges. $\diamond$

Theorem 21 (Abel's Test). If the sum $\sum x_n y_n$ satisfies

- x_n monotone decrease
- $\sum y_n$ is convergent

Then the sum converges.

To convert Abel to Dirichlet, we can consider the sum $\sum (x_n - x)y_n$ and apply Dirichlet.

1.2 Series of Functions

Terminologies for convergence of $\{S_n(x)\}$ is used for $\sum_{n=1}^{\infty} f_n(x)$.

We can talk about absolute pointwise convergence, pointwise uniform, uniform pointwise, etc.

Proposition 5 (Cauchy Criterion). $\sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent on E iff for any $\varepsilon > 0$ there is M_ε such that

$$\|f_{m+1}(x) + \dots + f_n(x)\| < \varepsilon \quad \text{if } M_\varepsilon \leq m < n.$$

Theorem 22 (Weierstrass M-Test). Suppose there exists a sequence of numbers $M_n \geq 0$ such that

$$\|f_n\|_\infty \leq M_n.$$

and

$$\sum M_n < \infty.$$

Then $\sum f_n(x)$ converges absolutely uniformly on E .

Proof.

$$\begin{aligned} & \|f_{m+1}(x) + \dots + f_n(x)\| \\ & \leq \|f_{m+1}(x)\| + \dots + \|f_n(x)\| \\ & \leq M_{m+1} + \dots + M_n < \varepsilon \quad \text{if } K_\varepsilon \leq m < n. \end{aligned}$$

□

Theorem 23 (Dirichlet Test). Consider the series $\sum_{n=1}^{\infty} a_n(x)b_n$. If

- b_n monotone decrease and $\rightarrow 0+$
- $S_n(x)$ uniformly bounded on E

Then the series converges uniformly.

Proof. Same proof as for numbers.

□

Lecture 36

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Theorem 24. (Uniform convergence and integrability) Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are R-S integrable wrt g , g monotone increase on $[a, b]$. If $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent on $[a, b]$ then f is R-S integrable wrt g on $[a, b]$ and

$$\int_a^b f dg = \sum_{n=1}^{\infty} \int_a^b f_n dg.$$

Theorem 25. (Uniform convergence and Differentiability) Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ (continuously) differentiable on $[a, b]$ such that

- $\sum_{n=1}^{\infty} f'_n(x)$ is uniformly convergent on $[a, b]$
- $\sum_{n=1}^{\infty} f_n(x_0)$ is convergent for some $x_0 \in [a, b]$.

Then, $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is convergent uniformly on $[a, b]$, and $f'(x) = \sum_{n=1}^{\infty} f'_n(x)$, $x \in [a, b]$.

This justifies term-by-term differentiation.

Remark. This theorem is true without assuming continuously differentiable. However the proof we give will depend on this assumption.

Proof. From assumptions and Newton Leibniz formula, we have

$$f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) dt.$$

Taking sums,

$$\sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} f_n(x_0) + \sum_{n=1}^{\infty} \int_{x_0}^x f'_n(t) dt.$$

Applying definition of $f(x)$ and uniform convergence, we have

$$f(x) = \sum_{n=1}^{\infty} f_n(x_0) + \int_{x_0}^x \sum_{n=1}^{\infty} f'_n(t) dt =: f(x_0) + \int_{x_0}^x g(t) dt.$$

By the differentiation theorem,

$$f'(x) = g(x) \quad \text{for any } x \in [a, b].$$

□

Example. Show that

$$f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2} \quad x \in \mathbb{R}$$

where

$$(\alpha) = \alpha - [\alpha] \in [0, 1)$$

is the fractional part of α , is discontinuous at all rational points, but is Riemann integrable on any finite interval. ◇

By Weierstrass M-test, $\sum f_n(x)$ is uniformly convergent.

f_n has finitely many discontinuity, so f_n is Riemann integrable, hence f is Riemann integrable.

f_n not continuous at all c , but the one sided limits exist at every point. Define

$$j_n(c) = f_n(c+) - f_n(c-).$$

be the *jump* at c .

By uniform continuity,

$$f(c+) = \lim_{x \rightarrow c+} \sum f_n(x) = \sum \lim_{x \rightarrow c+} f_n(x).$$

Similarly we define $f(c-)$. Now we have

$$f(c+) - f(c-) = \sum_{n=1}^{\infty} j_n(c) < 0 \Leftrightarrow c \in \mathbb{Q}.$$

1.3 Power Series

A power series is a series of the form

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Theorem 26. Let $\rho = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \in [0, \infty]$ and $R = \frac{1}{\rho}$ in the extended sense.

Then

1. for any $|x| < R$ the series is convergent.
2. for any $|x| \leq C < R$ the series is uniformly convergent.
3. If $|x| > R$ the series is divergent.
4. for $|x| = R$ the series may or may not diverge.

R is called [\[\[Radius of Convergence \]\]](#) .

Proof. Apply Root test to $a_n x^n$. Thus

$$\limsup |a_n x^n|^{\frac{1}{n}} = |x| \limsup |a_n|^{\frac{1}{n}} = \frac{|x|}{R} < 1 \Leftrightarrow |x| < R.$$

To prove the uniformly convergent part, let $x \in [-r, r]$ and

$$|a_n x^n| \leq |a_n| r^n.$$

and

$$\sum |a_n| r^n < \infty \quad \text{by Root Test.}$$

Hence $\sum a_n x^n$ uniformly converge by Weierstrass M-test. $\square$

Theorem 27. If f is a power series as above, and R is as above, then f is infinitely differentiable in $(-R, R)$ and the series can be differentiated term-by-term.

Proof. Consider the series $g(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$. $g(x)$ converges iff $xg(x) = \sum_{k=0}^{\infty} k a_k x^k$ converges. We claim this sum has the same radius of convergence as g . This is true since $k^{\frac{1}{k}} \rightarrow 1$.

Besides, we know f is convergent at 0. Apply differentiation theorem, $f'(x) = \sum_{k=0}^{\infty} k a_k x^{k-1}$. $\square$

Remark. From term-of-term differentiation we have

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

and $f(x)$ is a [\[\[Taylor series \]\]](#).